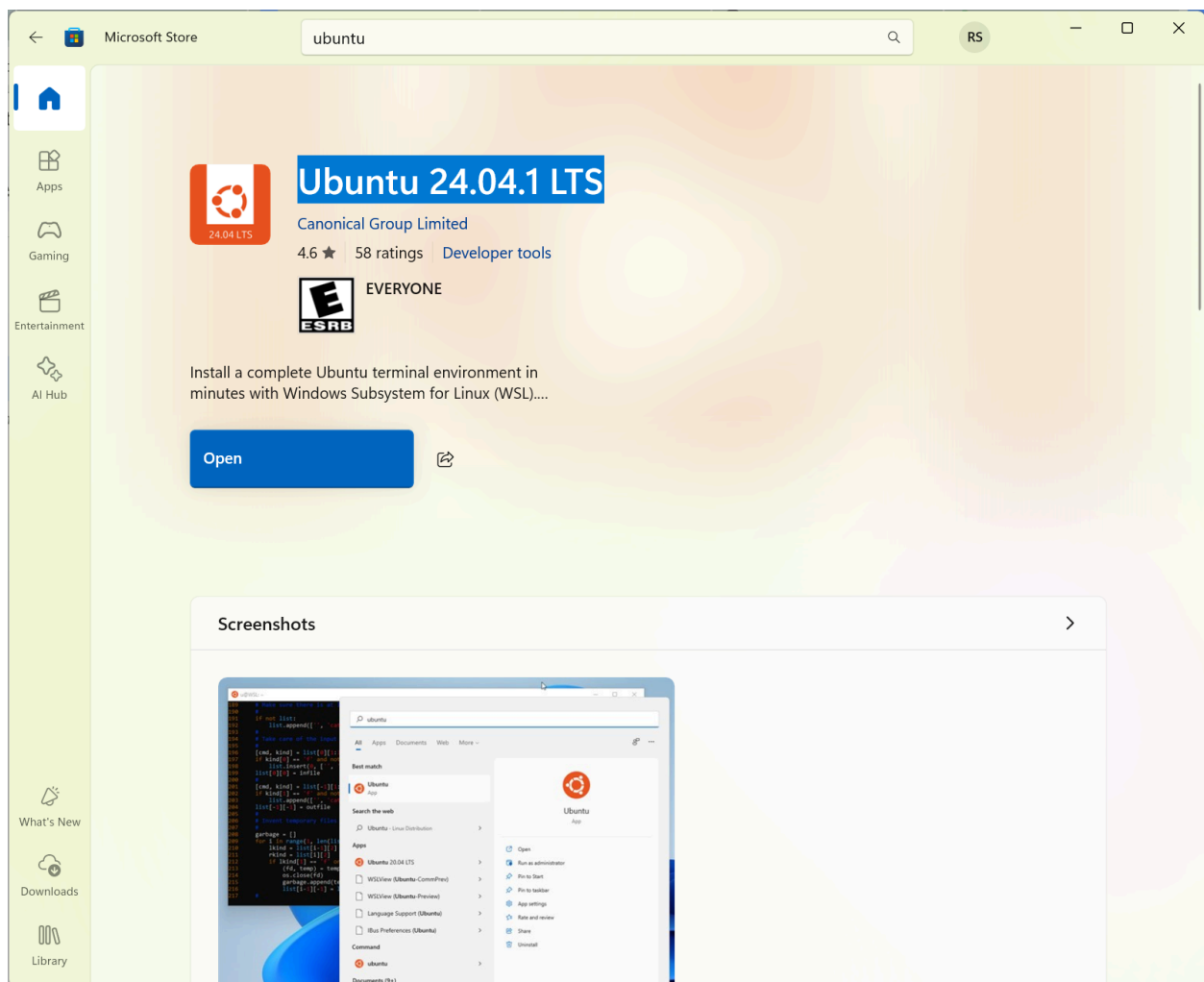


This is a step by step process to install all of the software necessary to read the CEFI data from the NOAA “o” Data Distribution archive of the CEFI MOM6 Regional Model output.

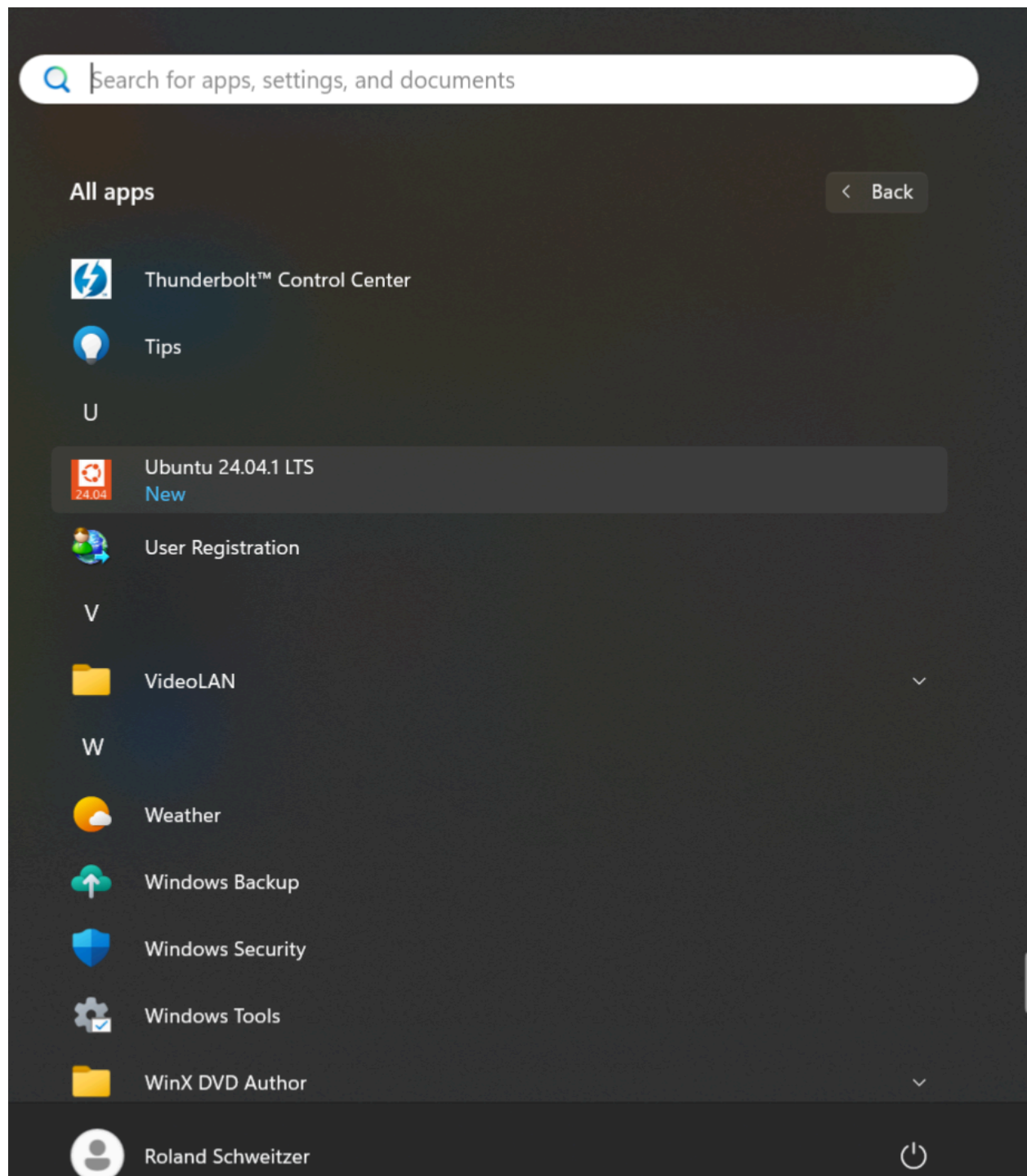
The environment used was Windows Subsystem for Linux on a private (non-government) Windows 11 PC. A similar procedure should work for any Linux environment.

Prerequisites:

Install a distribution of Ubuntu (or similar). I used Ubuntu 24.04.1 LTS which I installed from the Windows Store.



I used Windows Terminal to run the WSL distribution, you can also run it by starting it from you list of Windows applications.



Step 1 - Install a Python Package Manager

Install a Python package manager. I like mamba. You can install the latest release from [here](#).

```
wget
https://github.com/conda-forge/miniforge/releases/download/23.3.1-0/Miniforge3-23.3.1-0-Linux-x86_64.sh
sh ./Miniforge3-23.3.1-0-Linux-x86_64.sh
```

Step 2 - Create and install the environment

Create and install the software environment necessary to run the notebook. You can do this by downloading the environment.yml file from the github and running the following command (after you have started a new terminal session and mamba is activated. See (base) in front of you Linux command prompt).

```
(base) rhs@ThinkPad:~$ wget
https://raw.githubusercontent.com/NOAA-PMEL/GoA_xpublish/refs/heads/main/read_nwa/environment.yml
(base) rhs@ThinkPad:~$ mamba env create -f environment.yml
```

When successful, this command will install an environment called “nwa”

Step 3 - download the kerchunk index.

This file contains the aggregated index for all of the time steps in this forecast. You can use that one file to access all the variables and all the times in the data set. The notebook illustrates how to slice the data along the time axis.

```
wget
https://noaa-oar-cefi-regional-mom6-nwa-pds.s3.amazonaws.com/northwest_atlantic/full_domain/seasonal_reforecast/monthly/regrid/r20240213/all.json
```

Step 4 - Get and run the notebook

To run the notebook you have to download it, activate the mamba environment you just create, and run jupyter lab.

```
wget
https://raw.githubusercontent.com/NOAA-PMEL/GoA_xpublish/refs/heads/main/read_nwa/read_nwa_seasonal_reforecast_monthly_regird_s3.ipynb
```

```
mamba activate nwa  
jupyter lab
```

Control-click the offered link (the file:// link should work) and run the notebook.